

How a Supervised Learning Model Works

The core process of supervised learning involves taking a dataset, feeding it into a learning algorithm, and producing a predictive model.

1. **Input (Training Set):** The process begins with a training set containing both the input features (x) and the correct output targets (y).
 2. **Learning Algorithm:** The training set is fed into a learning algorithm.
 3. **Output (The Model):** The learning algorithm's output is a **model**, which is a function, denoted as f .
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The Model's Job: Making Predictions 🎱

- The model f takes a new input x and produces an estimated output, called a **prediction**.
 - **Prediction Notation:** The prediction is denoted by $\hat{y}$ (pronounced "y-hat").
 - y represents the **true value** or target (from the training data).
 - $\hat{y}$ represents the model's **estimated value** or prediction.
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Representing the Model: Linear Regression

A key step in designing a model is choosing how to represent the function f . For linear regression, we use the equation of a straight line.

- **The Model Function:**
$$f_{w,b}(x) = wx + b$$
 - This is often shortened to $f(x) = wx + b$.
 - x is the input feature (e.g., house size).
 - w and b are the **parameters** of the model. w is the weight (slope of the line) and b is the bias (y-intercept). The learning algorithm's job is to find the best values for w and b to fit the data.
- **Why a Linear Model?** While more complex, non-linear functions (curves) exist, a linear model is a simple and effective foundation for understanding more complex models later.
- **Model Name:** This specific model is called **linear regression with one variable**, or **univariate linear regression**, because it uses a single input variable (x).